

Data Science Fundamentals with Python Learners' Space 2026

Worksheet 1

We hope you had a good time watching the tutorials for the course and have learned a lot from them. Let's try out the following questions!

Question 1:

Create a Jupyter notebook or Google Colab notebook. Initialize and print a 2-dimensional NumPy array of shape (5, 5) filled with random* integers between 1 and 100. Perform the following tasks:

- Extract and print the middle element of the array using NumPy indexing.
- Calculate and print the mean of each row of the array.
- Create a new array that contains only the elements from the original array that are greater than the overall mean of the array.
- Write a Python function `numpy_spiral_order(matrix)` that takes a NumPy matrix and returns a list containing the elements visited in a spiral order. For example:



```
arr = np.random.randint(1, 100, (5, 5))
print(arr)
row_means = np.mean(arr, axis=1)
print(row_means)
spiral_order = numpy_spiral_order(arr)
print(spiral_order)
```

[[4 7 9 6 5]
[8 1 3 2 10]
[12 13 14 15 16]
[17 18 19 20 21]
[22 23 24 25 26]]

row_means: [5. 5.5 13. 18. 20.]

Spiral order: [1, 2, 3, 4, 5, 16, 15, 14, 13, 12, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 6, 7, 8, 9, 10, 11, 25, 24, 23]

Figure 1: Output of `numpy_spiral_order`

Hints:

- Use `np.random.randint()` to create the initial array `arr`.
- Use NumPy functions like `np.mean()` and boolean indexing to perform calculations and extract elements.

Question 2:

This question involves the application of Python libraries on a dataset of video game sales. Here is the link to the [dataset](#).

Create a Jupyter Notebook or a Google Colab Notebook, download the dataset in the drive link, and solve the following questions using the required Python libraries:

- (a) Add a column of 'global sales' showing the total sales of all the different regions to the data frame and sort (highest first) and print the DataFrame according to it.
- (b) Display a plot of the total number of copies sold of each genre globally.
- (c) Filter out only the games containing 'Grand Theft Auto' in their name and display the following as a DataFrame:
 - i. their name
 - ii. the platform they were released on
 - iii. the year they were released in
 - iv. the sum of sales in only Europe and Japan
- (d) Display a pie chart of the total sales of all Grand Theft Auto games combined in North America, Europe, Japan, and other sales.